

[106]

✓ Os

```
df.filter(col("Heart Rate (bpm)" > 100).show(20, truncate=False)
```

Patient Number	Heart Rate (bpm)	SpO2 Level (%)	Systolic Blood Pressure (mmHg)	Diastolic Blood Pressure (mmHg)	Body Temperature (°C)	Fall Detection	Predicted Disease	Data
4	137	84	167	97	37.0187665357031	Yes	Asthma	99
6	123	88	110	99	37.33092741815	No	Hypertension	94
8	103	93	112	71	37.7473440741132	Yes	Asthma	80
9	111	92	102	99	36.4721515117444	Yes	Asthma	82
10	146	90	176	85	36.6235619295666	Yes	Arrhythmia	90
11	135	95	128	67	36.6567420281208	No	Hypertension	94
13	102	88	117	68	36.8625355600735	Yes	Diabetes Mellitus	83
15	140	93	128	68	37.7132798559411	No	Hypertension	92
16	116	91	134	96	37.6730668015071	Yes	Diabetes Mellitus	94
18	137	84	102	87	37.6867138769215	Yes	Arrhythmia	93
20	117	92	141	90	37.9980797905673	No	Normal	89
21	140	86	104	88	37.124229295728	No	Arrhythmia	80
22	134	85	143	78	36.2338963668331	Yes	Normal	96
23	106	81	120	69	36.5693945949311	Yes	Asthma	85
26	110	92	175	95	37.6819763627835	No	Asthma	86
27	122	99	125	95	37.5808698546241	No	Hypertension	90
32	102	96	167	99	36.1706224534299	No	Hypertension	80
33	133	86	137	88	37.5731245799956	No	Hypertension	80
35	108	83	146	99	36.2488595998062	Yes	Hypertension	94
36	121	85	165	70	36.7987329527629	No	Hypertension	87

[108]

✓ Os

```
from pyspark.sql.functions import col
```

```
filtered_batch_windows = batch_windows.filter(col("avg_hr") > 100)  
filtered_batch_windows.show(20, truncate=False)
```

window	Patient Number	avg_hr
{2026-05-28 22:14:00, 2026-05-28 22:16:00}	39406	149.0
{2026-06-02 06:14:00, 2026-06-02 06:16:00}	6302	149.0
{2026-05-06 14:14:00, 2026-05-06 14:16:00}	49942	149.0
{2026-05-31 09:14:00, 2026-05-31 09:16:00}	24347	149.0
{2026-06-03 11:14:00, 2026-06-03 11:16:00}	3273	149.0
{2026-06-09 19:14:00, 2026-06-09 19:16:00}	28121	149.0
{2026-06-04 03:14:00, 2026-06-04 03:16:00}	4257	149.0
{2026-05-16 10:14:00, 2026-05-16 10:16:00}	36706	149.0
{2026-06-05 04:14:00, 2026-06-05 04:16:00}	5232	149.0
{2026-05-19 22:14:00, 2026-05-19 22:16:00}	15622	149.0
{2026-06-08 10:14:00, 2026-06-08 10:16:00}	21154	149.0
{2026-05-26 20:14:00, 2026-05-26 20:16:00}	2456	149.0
{2026-06-04 17:14:00, 2026-06-04 17:16:00}	39243	149.0
{2026-06-06 20:14:00, 2026-06-06 20:16:00}	40192	149.0
{2026-06-05 23:14:00, 2026-06-05 23:16:00}	21213	149.0
{2026-05-05 00:14:00, 2026-05-05 00:16:00}	25980	149.0
{2026-06-05 18:14:00, 2026-06-05 18:16:00}	2218	149.0
{2026-06-07 06:14:00, 2026-06-07 06:16:00}	46182	149.0
{2026-05-08 00:14:00, 2026-05-08 00:16:00}	10908	149.0
{2026-05-12 13:14:00, 2026-05-12 13:16:00}	17799	149.0

[144]

✓ 1s

```
windowed_batch.orderBy(desc("avg_heart_rate")).show(20, truncate=False)
```

▼

Patient Number	window_start	window_end	avg_heart_rate
48599	2026-05-20 22:18:00	2026-05-20 22:20:00	149.0
37276	2026-06-03 09:18:00	2026-06-03 09:20:00	149.0
11273	2026-06-03 12:18:00	2026-06-03 12:20:00	149.0
38070	2026-06-11 23:18:00	2026-06-11 23:20:00	149.0
33193	2026-06-06 20:18:00	2026-06-06 20:20:00	149.0
27922	2026-05-07 11:18:00	2026-05-07 11:20:00	149.0
16034	2026-06-13 11:18:00	2026-06-13 11:20:00	149.0
32493	2026-05-25 08:18:00	2026-05-25 08:20:00	149.0
14033	2026-06-13 12:18:00	2026-06-13 12:20:00	149.0
22026	2026-06-13 19:18:00	2026-06-13 19:20:00	149.0
18274	2026-06-03 11:18:00	2026-06-03 11:20:00	149.0
14742	2026-05-14 23:18:00	2026-05-14 23:20:00	149.0
43874	2026-05-09 11:18:00	2026-05-09 11:20:00	149.0
7608	2026-05-20 13:18:00	2026-05-20 13:20:00	149.0
9368	2026-05-30 13:18:00	2026-05-30 13:20:00	149.0
31966	2026-05-05 15:18:00	2026-05-05 15:20:00	149.0
29270	2026-06-03 15:18:00	2026-06-03 15:20:00	149.0
45651	2026-05-18 18:18:00	2026-05-18 18:20:00	149.0
45878	2026-05-09 07:18:00	2026-05-09 07:20:00	149.0
2991	2026-05-04 14:18:00	2026-05-04 14:20:00	149.0

only showing top 20 rows

[150]

✓ 0s



```
windowed_batch.filter(  
    col("Patient Number") == 48599  
)orderBy("window_start").show(truncate=False)
```

▼

...

Patient Number	window_start	window_end	avg_heart_rate
48599	2026-05-20 22:24:00	2026-05-20 22:26:00	149.0

=====

CLINICAL ALERTS

Sustained Elevated HR (>100 bpm)

Across Two Consecutive Windows

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Patient Number	window_start	window_end	avg_heart_rate
48599	2026-05-20 22:24:00	2026-05-20 22:26:00	149.0

Total Clinical Alerts Generated: 1